

Dynamic Macroeconomics II

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Fiscal Policy: Ricardian Equivalence Model with Government

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Model with government

- Now, we add a government that purchases a stream of public expenditure $\{g_t\}_{t=0}^{\infty}$.
- Public expenditures do not appear in consumer's preferences. It is not productive. E.g. is is not public investment.
- The government levies lump-sum taxes $\{\tau_t\}_{t=0}^{\infty}$ on the household.
- Sequential budget constraint for the government is given by:

$$B_t + g_t = \tau_t + \frac{B_{t+1}}{R}$$

- B_t is one-period debt due at period t , denominated in units of the consumption good.
- $B_t > 0$ is public debt (notation is opposite compared to consumer).

- Solve forward for B_t using the government's budget constraint to get:

$$B_t = \sum_{j=0}^{\infty} R^{-j} (\tau_{t+j} - g_{t+j})$$

We did this for the consumer. **Check** you can do it here.

- To get that expression we used the Transversality Condition:

$$\lim_{T \rightarrow \infty} R^{-T} B_{t+T} = 0$$

Changes in consumer's problem

- Consumer's sequential budget constraint is modified to include taxes:

$$c_t + \frac{b_{t+1}}{R} \leq y_t + b_t - \tau_t$$

- Again, solve for b_t iterating forward, and use a Transversality Condition:

$$b_t = \sum_{j=0}^{\infty} R^{-j} (c_{t+j} + \tau_{t+j} - y_{t+j})$$

- Set $c_t = 0$ and obtain the new NDL:

$$\tilde{b}_t = - \sum_{j=0}^{\infty} R^{-j} (y_{t+j} - \tau_{t+j})$$

- By making consumption as small as possible the consumer can borrow and repay at most the present value of its **disposable** income

An equilibrium for this economy is a plan for the consumer $\{c_t, b_{t+1}\}_{t=0}^{\infty}$ and a government policy $\{g_t, \tau_t, B_{t+1}\}_{t=0}^{\infty}$ such that, given initial conditions (b_0, B_0) :

- 1 The consumer solves her problem given the sequence of taxes.
- 2 The government satisfies its budget constraint every period.

- Does the timing of taxes matter?
- The general idea is that, for example, lower taxes today imply more debt today, and higher taxes in the future to pay the debt.
- The consumer knows this and its consumption plan is not altered
- This is true if her **intertemporal** budget constraint does not change with the alternative sequence of taxes.

Ricardian Proposition

- Suppose we impose the NDL.
- Given initial conditions (b_0, B_0) , let $\{c_t, b_{t+1}\}_{t=0}^{\infty}$ and $\{g_t, \tau_t, B_{t+1}\}_{t=0}^{\infty}$ be an equilibrium.
- Consider any other tax sequence $\{\bar{\tau}_t\}_{t=0}^{\infty}$ such that:

$$\sum_{t=0}^{\infty} R^{-t} \tau_t = \sum_{t=0}^{\infty} R^{-t} \bar{\tau}_t$$

- Then, $\{c_t, \bar{b}_{t+1}\}_{t=0}^{\infty}$ and $\{g_t, \bar{\tau}_t, \bar{B}_{t+1}\}_{t=0}^{\infty}$ also constitute an equilibrium, in which :

$$\bar{b}_t = \sum_{j=0}^{\infty} R^{-j} (c_{t+j} + \bar{\tau}_{t+j} - y_{t+j})$$

$$\bar{B}_t = \sum_{j=0}^{\infty} R^{-j} (\bar{\tau}_{t+j} - g_{t+j})$$

- Step 1: need to show that under the new tax scheme, household chooses same consumption stream as before.
- For any tax plan, from seq. constraint at $t = 0$ solve forward for b_0 :

$$b_0 = \sum_{t=0}^{\infty} R^{-t}(c_t - y_t) + \sum_{t=0}^{\infty} R^{-t}\tau_t$$

- Optimal plan for consumption does not depend on timing of taxes, only on their present value.
- Since present value of new tax plan is the same as originally, consumer's intertemporal budget constraint is unaltered
- Therefore she chooses same plan.

$$c + b \leq y + (1+r)b \Rightarrow c + b/(1+r) \leq y + b$$

- Step 2: construct adjusted borrowing plan.
- From seq. constraint in t solve forward for $\bar{b}_t$ using new tax sequence and same consumption plan to get:

$$\bar{b}_t = \sum_{j=0}^{\infty} R^{-j} (c_{t+j} + \bar{\tau}_{t+j} - y_{t+j})$$

- Note that we assumed this new sequence satisfies a Transv. Cond. We will prove it later.

- Step 3: show new debt sequence satisfies NDL.
- Question: $\bar{b}_t \geq \tilde{b}_t$?
- It is easy to show that the above sequence satisfies the natural debt limit, since the consumption stream $\{c_t\}_{t=0}^{\infty}$ is non-negative.
- We have

$$\bar{b}_t = \sum_{j=0}^{\infty} R^{-j} (c_{t+j} + \bar{\tau}_{t+j} - y_{t+j}) \geq \tilde{b}_t = - \sum_{j=0}^{\infty} R^{-j} (y_{t+j} - \bar{\tau}_{t+j})$$

- Step 4: show that Transversality Condition holds.
- Consider any period $k - 1 \geq 0$ and **solve backward for $\bar{b}_k$** to get (**check you can do it**):

$$\bar{b}_k = \sum_{j=1}^k R^j [y_{k-j} - \bar{\tau}_{k-j} - c_{k-j}] + R^k b_0$$

- Note that b_0 is the same in both asset sequences, it is exogenous.
- Now, call b_k the value of the above equation for the original tax plan $\{\tau_t\}_{t=0}^{\infty}$.
- Compute the difference between those two values as:

$$b_k - \bar{b}_k = \sum_{j=1}^k R^j [\bar{\tau}_{k-j} - \tau_{k-j}]$$

- Multiply both sides by R^{1-k} . The previous equality can be written as (**check**)

$$R^{1-k}(b_k - \bar{b}_k) = R \sum_{t=0}^{k-1} R^{-t} [\bar{\tau}_t - \tau_t]$$

- Then, when $k \rightarrow \infty$, if borrowing plan $\{b_{t+1}\}_{t=0}^{\infty}$ satisfies transversality condition, so must $\{\bar{b}_{t+1}\}_{t=0}^{\infty}$.
- We are done with the consumer. Now comes the government.

- Step 5: Need to show that with new taxes government satisfies budget constraint
- Budget constraint at $t = 0$ does not depend on timing of taxes, only on their present value. Solve forward for B_0 :

$$B_0 = \sum_{t=0}^{\infty} R^{-t} \tau_t - \sum_{t=0}^{\infty} R^{-t} g_t$$

- With an unchanged present value of taxes, the government can finance the same expenditure plan $\{g_t\}_{t=0}^{\infty}$.

Q.E.D.